

**MULTIMODAL DOCUMENT
INTELLIGENCE SYSTEM FOR
INVOICE AND RECEIPT PROCESSING**



**USING VISION-LANGUAGE MODELS WITH LAYOUT-
AWARE OCR**

A PROJECT REPORT

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We affirm that the project work titled “**MULTIMODEL DOCUMENT INTELLIGENCE SYSTEM FOR INVOICE AND RECEIPT PROCESSING USING VISION-LANGUAGE MODELS WITH LAYOUT-AWARE OCR**” being submitted in partial fulfillment for the award of Bachelor of Engineering degree is the original work carried out by us. It has not formed the part of any other project work submitted forward of any degree or diploma, either in this or any university.

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ABSTRACT

In the modern business environment, extracting structured data from commercial documents such as invoices and point-of-sale receipts is a critical challenge. Traditional Optical Character Recognition (OCR) systems systematically fail on real-world inputs due to non-standardized layouts, physical paper degradation, and widespread code-switching across Indian regional scripts. Legacy pipelines depend on sequential text box detection and text recognition, causing early bounding-box errors to propagate downstream, requiring extensive manual entry. To overcome these limitations, this project introduces a Multimodal Document Intelligence System (MDIS) operating on an OCR-free, pixel-level visual extraction paradigm. The system is driven by the Qwen2.5-VL-32B Vision-Language Model (VLM), which bypasses intermediate OCR text steps by directly interpreting spatial layouts and native script elements at the pixel level. A localized Contrast Limited Adaptive Histogram Equalization (CLAHE) stage restores faded thermal ink, while an aspect-ratio preserving vertical overlap splitting algorithm processes tall document slices within visual token capacities. To eliminate local computing constraints, a distributed Master-Worker architecture is implemented. A lightweight FastAPI master node handles local preprocessing and serialized prompt packaging. Encrypted Cloudflare Tunnels securely route payloads to remote dual NVIDIA T4 GPU workers running the 32B model with IQ4 XS 4-bit quantisation for high-speed inference. A Chain-of-Thought (CoT) dual-prompt extracts regional characters and parallel English translation fields in a single forward pass, while a monospace ASCII grid "Digital Twin" output provides spatial document layout reconstruction for immediate human auditing. On a diverse Indic financial document corpus, the proposed system achieves a Key-Value Accuracy of 96% on invoices and 92% on degraded receipts. This approach outpaces traditional OCR-based baselines, significantly decreases transaction latency, and delivers a scalable, cost-effective document parsing system for automated business accounting workflows.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	DEFINITION
OpenCV	Open-Source Computer Vision Library
JSON	JavaScript Object Notation
VLM	Vision-Language Model
OCR	Optical Character Recognition
API	Application Programming Interface
NLP	Natural Language Processing
CLAHE	Contrast Limited Adaptive Histogram Equalization
GGUF	GPT-Generated Unified Format
VRAM	Video Random-Access Memory
GPU	Graphics Processing Unit
SSE	Server-Sent Events

CHAPTER 1

INTRODUCTION

In modern enterprises, automated document processing plays a critical role in accounting, auditing, and supply chain operations. However, real-world documents, especially in the Indian context, are highly unstructured and often contain multilingual text, inconsistent formatting, and degraded visual quality

Traditional Optical Character Recognition (OCR) systems attempt to solve this problem using a two-stage pipeline: first, detecting text regions through bounding box localization, and second, recognizing the textual content within those regions. While this approach works reasonably well for clean, well-formatted documents, it struggles in real-world conditions. Errors in text detections such as missing or incorrectly segmented regions—propagate to the recognition stage, resulting in incorrect or incomplete outputs. Furthermore, OCR systems lack an inherent understanding of document structure and context. They often fail to preserve relationships between key elements, such as associating item names with their corresponding prices in a receipt, leading to loss of semantic meaning.

To overcome these challenges, this work proposes a Vision-Language Model (VLM)-based system that fundamentally redefines the document processing pipeline. Instead of relying on intermediate text extraction steps, the proposed approach directly interprets document images at the pixel level and generates structured outputs in formats such as JSON. By leveraging advanced multimodal learning, the system simultaneously captures both textual content and spatial relationships within the document. This enables a deeper contextual understanding, allowing the model to accurately identify key-value pairs, tabular data, and hierarchical structures without explicit layout rules.

1.1 OBJECTIVE

The main objective of this research is to develop an efficient and intelligent **Multimodal Document Intelligence System** capable of accurately extracting structured information from complex, real-world financial documents such as invoices and receipts. The system aims to overcome the limitations of traditional OCR-based approaches by leveraging advanced Vision-Language Models (VLMs) for direct image understanding and structured data generation.

1.2 AIM

The aim of this project is to design and implement an advanced Multimodal Document Intelligence System that can efficiently process and understand complex financial documents such as invoices and receipts. The system aims to utilize Vision-Language Models (VLMs) to directly interpret document images and generate structured outputs without relying on traditional OCR pipelines. It focuses on achieving high accuracy, layout awareness, and multilingual understanding, enabling reliable extraction of key information from real-world, unstructured documents. The system is intended to reduce manual data entry, improve processing speed, and provide a scalable, cost-effective solution for enterprise document automation.

CHAPTER 2

LITERATURE SURVEY

1. Vision Transformer for Image Understanding – Dosovitskiy et al. (2021).

This study introduced Vision Transformer (ViT), which processes images as patch sequences instead of using traditional convolution methods. It helps models understand visual patterns and spatial relationships in images. This concept supports our system because invoices and receipts need layout-aware visual understanding [1].

2. Attention Mechanism – Vaswani et al. (2017).

This paper proposed the Transformer architecture using self-attention. Self-attention helps models focus on important parts of input data. In our project, attention-based visual understanding helps connect related fields such as item names, prices, subtotal, tax, and total amount [2].

3. Qwen-VL: Vision-Language Model – Qwen Team (2023).

Qwen-VL is a vision-language model capable of understanding images and text together. It can read document images, recognize layout, and generate structured responses. This is highly relevant to our project because we use VLM-based extraction instead of depending only on OCR [3].

4. LayoutLM for Document Understanding – Xu et al. (2020).

This LayoutLM combines text and layout information for document image understanding. It improves extraction from forms and invoices by considering word positions. However, it still depends on OCR output, so OCR errors can affect final results. [4].

5. Donut: OCR-Free Document Understanding – Kim et al. (2022).

Donut introduced an OCR-free approach for document understanding by directly converting document images into structured text. It reduces dependency on OCR engines. But it may struggle with high-resolution variations and complex layouts, which our VLM-based system aims to handle better [5].

6. Adaptive Histogram Equalization – Pizer et al. (1987).

This study introduced adaptive histogram equalization techniques to improve image contrast. In our project, CLAHE is used to enhance faded receipts and improve visibility before sending the image to the model. This helps improve extraction accuracy from low-quality documents [6].

7. TextMonkey: An OCR-Free Large Multimodal Model – Liu et al. (2024)

This paper introduces TextMonkey, a large multimodal model tailored for document text understanding. By using split image patches and token resampling, it achieves high-performance OCR-free text extraction and visual grounding on complex documents, which aligns with our spatial layout understanding goals [7].

8. Tesseract OCR Engine - Smith, R. (2007).

Tesseract is one of the most widely used OCR engines for text extraction from images. It performs well on clean scanned documents but struggles with faded thermal receipts, multilingual text, overlapping stamps, and irregular layouts. These limitations motivate our VLM-based approach [8].

9. LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking – Huang et al. (2022).

This paper presents LayoutLMv3 improves document AI by combining text and image masking. It performs well in form understanding, receipt understanding, and document layout analysis [9].

10. PP-OCR: A Practical Ultra Lightweight OCR System – Du, Y. et al. (2020).

PaddleOCR 3.0 focuses on multilingual OCR, document parsing, and key information extraction. It is useful for invoice and receipt processing, but still follows an OCR-based pipeline [11].

11. Universal Document Processing (UDOP) – Tang et al. (2023)

This paper introduces UDOP, a unifying framework that integrates vision, text, and layout for universal document processing. It uses a sequence-to-sequence model to jointly perform document content generation and layout reconstruction, showcasing strong capabilities in form understanding and document parsing [12].

12. End-to-End Document Recognition – Davis et al. (2022).

This study presents Dessurt, an end-to-end model for document recognition and understanding. It processes document images directly using a transformer architecture to generate text, bypassing traditional layout analysis and OCR binarisation steps to improve structured data extraction [13].

13. Pix2Struct: Screenshot Parsing – Lee et al. (2023).

This research introduces Pix2Struct, a visual language model pre-trained by parsing screenshots into HTML structure. It demonstrates excellent performance on visually rich document understanding and web-page parsing tasks by learning to map screenshot pixels directly to structured text representations [14].

14. Chain-of-Thought Prompting in LLMs – Wei et al. (2022)

This work introduces Chain-of-Thought (CoT) prompting, showing that eliciting multi-step reasoning pathways in large language models significantly improves performance on complex reasoning tasks. In our system, a CoT dual-prompt is used to extract native script text and translate it to English in a single forward pass [15].

CHAPTER 3

SYSTEM STUDY

3.1 EXISTING SYSTEM

- **OCR-Based Systems (Tesseract, EasyOCR)**

In OCR (Optical Character Recognition) systems are widely used for extracting text from document images. These systems follow a two-step process involving text detection and recognition. They work well for clean and structured documents but perform poorly when dealing with noisy images, faded receipts, and complex layouts. Additionally, OCR systems extract only raw text and fail to preserve the spatial relationships between different elements in the document. Existing OCR-based systems do not consider the current system.

- **Rule-Based Parsing Methods**

Rule-based systems extract specific fields using predefined patterns and keywords such as “Total”, “Date”, or “Invoice Number”. While they are simple to implement, they require manual configuration for each document format. These systems are not scalable and fail when document structure changes, making them unsuitable for handling diverse real-world documents.

- **Inefficient Deployment on Edge Devices**

Existing mode systems do not consider the spatial arrangement of text within a document. They fail to recognize the relationship between different sections such as headers, tables, and line items. Since invoices and receipts often follow semi-structured formats, the absence of layout awareness leads to incorrect mapping of values and reduces overall extraction accuracy.

- **Manual Data Entry Requirement**

Due to the limitations of OCR systems, extracted data often requires manual verification and correction. Users must manually identify and input key details such as invoice number, date, and total amount. This increases processing time, introduces human errors, and reduces efficiency in large-scale document handling environments.

- **Poor Performance on Handwritten and Noisy Documents**

Traditional OCR systems perform poorly when handling handwritten receipts, low-quality scans, or images with noise, shadows, or distortions. Variations in handwriting styles and image quality significantly affect text recognition accuracy, making the system unreliable for practical use.

3.2 DISADVANTAGES OF EXISTING SYSTEM

- Entire system output depends on OCR quality; small errors lead to incorrect final extraction.
- Fails to handle faded receipts, blurred images, and low-quality scans effectively.
- Cannot preserve spatial relationships like item–price alignment in receipts.
- Struggles with irregular formats, overlapping text, and borderless tables.
- Performs poorly on documents containing mixed languages like English + Tamil/Hindi.
- Requires manual rules for each format, making it difficult to adapt to new document types.
- Extracts only text without understanding meaning or relationships between fields.

3.3 PROBLEM STATEMENT

At present, In real-world applications, invoices and receipts are not in a fixed format and often contain noise, faded text, and multiple languages. Traditional OCR-based systems try to extract text, but they fail when the document quality is poor or the layout is complex. These systems also cannot maintain the correct relationship between data like item and price. Rule-based methods are not flexible and fail when the format changes. Even advanced models depend on OCR, so errors still occur. Hence, a better system is needed to accurately understand documents and extract structured data directly.

3.4 PROPOSED SYSTEM

The proposed system uses a **Vision-Language Model (VLM)** to directly understand document images and extract structured data without using traditional OCR. It can handle

complex layouts, faded text, and multilingual content more effectively. The system captures both text and spatial relationships, ensuring accurate mapping of information like item and price.

It also generates structured output in JSON format along with a Digital Twin for easy verification. A Master-Worker architecture is used to perform heavy processing in the cloud while keeping the local system lightweight. This approach improves accuracy, reduces errors, and supports real-time document processing. The system includes the following key features:

- **OCR-Free Document Understanding**

The system directly processes document images using Vision-Language Models. This avoids OCR errors and improves overall extraction accuracy.

- **Layout-Aware Extraction**

The model understands both text and its position in the document. This helps in correctly mapping relationships like item and price.

- **Multilingual Support**

The system can handle documents with multiple languages. It accurately extracts and standardizes mixed-language content.

- **Digital Twin Representation**

Generates a layout-preserving text grid of the document. This makes manual verification simple and user-friendly.

3.5 ADVANTAGES OF PROPOSED SYSTEM

- Direct image understanding reduces errors compared to OCR-based systems.
- Supports mixed-language documents like English + Tamil/Hindi.
- Maintains spatial relationships such as item and price alignment.
- Automatically extracts structured data, minimizing human intervention.
- Can adapt to different document formats without rule changes.
- Uses GPU-based processing for faster and powerful computation.

CHAPTER 4

METHODOLOGY

4.1 EXISTING METHOD

The existing document processing systems are mainly based on traditional OCR and rule-based extraction approaches. These systems attempt to extract text from invoices and receipts using OCR engines and then apply predefined rules for identifying important fields. However, such systems are not efficient in handling complex document layouts, multilingual content, noisy images, and real-world unstructured financial documents. The overall workflow of the existing methods is explained below.

- **Data Acquisition**

In existing systems, invoice or receipt images are collected from scanners, mobile devices, or uploaded PDF files. These documents may contain noise, blur, faded text, handwritten content, or inconsistent layouts, which affect extraction quality.

- **OCR-Based Text Extraction**

Traditional systems use OCR engines such as Tesseract or PaddleOCR to extract raw text from the document image. While OCR helps in converting image content into text, the extracted output often contains recognition errors, especially in low-quality or multilingual documents.

- **Preprocessing**

Basic preprocessing techniques such as resizing, filtering, and noise removal are applied to improve image quality before OCR processing. However, many existing systems do not apply advanced enhancement techniques required for handling real-world invoice and receipt images effectively.

- **Rule-Based Parsing**

After OCR extraction, predefined rules and templates are used to identify fields like invoice number, vendor name, date, and total amount. These methods work only for fixed

document formats and fail when the layout changes or when documents contain irregular structures.

- **Layout Processing**

Some advanced systems use layout-aware models such as LayoutLM or LayoutLMv3 to understand document structure using OCR text and bounding box information. Although these models improve layout understanding, they still depend heavily on OCR quality, which leads to error propagation.

- **Data Extraction**

The processed text is analyzed to extract important details such as item names, prices, taxes, and totals. However, existing systems often struggle to maintain the correct relationships between fields, especially in tables and complex invoice layouts.

- **Limitations of Existing Systems**

Existing systems have high dependency on OCR accuracy and are not suitable for highly unstructured or multilingual documents. They require manual configuration, struggle with noisy images, and often fail to preserve semantic and spatial relationships within documents.

4.2 PROPOSED METHOD

To overcome the severe limitations of legacy OCR-based document parsing pipelines, this work proposes a Multimodal Document Intelligence System (MDIS) that operates on an OCR-free, pixel-level extraction paradigm. The core intelligence is driven by the Qwen2.5-VL-32B Vision-Language Model (VLM), which interprets the layout and text directly from visual features using a gated cross-attention transformer decoder. The proposed system features an advanced preprocessing workflow utilizing Contrast Limited Adaptive Histogram Equalization (CLAHE) for degraded receipts, a dynamic resolution-preserving overlap splitting algorithm for tall documents, and a Chain-of-Thought (CoT) dual-language prompt execution mechanism that extracts native script text and translated English fields simultaneously.

To bypass local hardware limitations, the proposed system employs a distributed Master-Worker architecture. A local, lightweight FastAPI server (Master Node) manages image acquisition and preprocessing, while secure Cloudflare Tunnels dynamically route quantised tensor calculations to remote dual NVIDIA T4 GPU workers (Worker Node). The final extraction yields structured, schema-validated JSON outputs alongside a monospace-aligned "Digital Twin" text-grid representation, which enables spatially faithful document visualization for human auditing without the need for frontend coordinate bounding-box rendering.

Extracted information is converted into structured **JSON format** and stored in **MongoDB** for efficient retrieval and management. This approach improves extraction accuracy, supports multilingual documents, and handles complex layouts more effectively than traditional systems.

4.2.1 Key Features of the Proposed Method

- **OCR-Free End-to-End Visual Extraction**

The system bypasses OCR engines entirely by processing raw document image patches. This eliminates character recognition and bounding box detection error propagation paths, substantially improving accuracy on skewed, noisy, or degraded inputs.

- **Contrast Limited Adaptive Histogram Equalization (CLAHE)**

Rather than relying on binarisation, which amplifies crease noise, the preprocessing pipeline applies localized adaptive contrast enhancement to make faded thermal inks highly legible before visual patch tokens are generated.

- **Dynamic Overlap Splitting Algorithm**

To prevent visual token budget overflow on tall thermal receipts, the system dynamically slices images along the vertical axis with an overlap margin (δ), guaranteeing that text lines intersecting chunk boundaries are captured in their entirety by at least one adjacent patch.

- **Gated Cross-Attention Multimodal Fusion**

The VLM architecture integrates visual patch embeddings directly into the layers of the 32-billion parameter language decoder through gated cross-attention. This enables the system to capture spatial key-value dependencies (such as row-to-price mapping in borderless tables) regardless of structural irregularities or overlapping merchant stamps.

- **Image Enhancement and Noise Removal**

Preprocessing techniques such as resizing and noise reduction improve image clarity before visual patch token generation. This helps the system handle blurred, faded, and low-quality receipt images effectively.

- **Chain-of-Thought (CoT) Dual-Language Translation**

The prompt execution layer utilizes a two-pass CoT strategy: the model is instructed to first transcribe the document exactly in its native regional script, and then output an aligned, English-translated JSON payload in a single forward pass, eliminating downstream API translation delays.

- **Spatially Faithful Digital Twin Verification**

The VLM is prompted to reconstruct the document as a monospace ASCII grid. This Digital Twin acts as a spatial layout representation, allowing human-in-the-loop auditors to instantly inspect alignment accuracy without navigating nested JSON trees.

CHAPTER 5

SYSTEM DESIGN

5.1 OVERALL SYSTEM ARCHITECTURE

The overall system architecture of the proposed Multimodal Document Intelligence System (MDIS) is designed as a distributed, zero-cost Master-Worker pipeline that handles visual extraction without local GPU hardware dependencies. The architecture comprises a lightweight local Master Node powered by FastAPI and React.js, and a high-performance remote Worker Node hosted on dual NVIDIA T4 GPUs. When a user uploads a multilingual document (such as a receipt or invoice) via the React dashboard, the image payload is captured by the local Master Node backend.

The Master Node applies local image preprocessing layers, including OpenCV-based Contrast Limited Adaptive Histogram Equalization (CLAHE) for localized binarisation enhancement, and a dynamic aspect-ratio vertical overlap splitting algorithm to chunk tall images without visual scaling distortions. The preprocessed image slices and task-specific prompts are serialized and transmitted through a secure, encrypted Cloudflare Tunnel bridge directly to the remote GPU Worker Node. The remote Worker Node executes quantized visual inference using the 32-billion parameter Qwen2.5-VL Vision-Language Model.

The model parses visual patch feature maps through spatial self-attention, outputting structured, schema-validated JSON data along with an aligned monospace ASCII grid "Digital Twin" visualization. The stitched and deduplicated payload is delivered back via Server-Sent Events (SSE) to the local client frontend for real-time dashboard display and auditing.

5.2 DATA FLOW DESIGN

The data flow design describes how information moves chronologically through the different layers of the proposed system. The workflow initiates when the user uploads an

invoice or thermal receipt image through the responsive React.js dashboard. The uploaded binary image is dispatched to the local FastAPI backend (Master Node), which initiates localized preprocessing. The Master Node first converts the document image to grayscale and applies Contrast Limited Adaptive Histogram Equalization (CLAHE) to restore faded textual areas, followed by dynamic vertical overlap splitting that segments tall receipts into overlapping image chunks with standard margin offsets (delta) to secure textual boundary continuity.

Next, the serialized image chunks and prompt templates are pushed through a secure Cloudflare Tunnel, bypassing local network firewalls. The remote Kaggle dual-T4 GPU worker node receives the data, deserializes the image chunks, and pipes them directly into the Qwen2.5-VL-32B Vision-Language Model. The visual patch encoder processes the visual features of the document directly at the pixel level, mapping spatial key-value associations through multi-head self-attention layers.

The language decoder executes a Chain-of-Thought (CoT) dual-prompt generation pass, transcribing regional scripts and outputting an English-translated schema-compliant JSON file parallel to a monospace ASCII "Digital Twin" text-grid. The local FastAPI Master Node captures this structured output, applies coordinate stitching and overlap boundary deduplication, and streams the validated results back to the React UI via Server-Sent Events (SSE) for human-in-the-loop review.

5.3 MODEL ARCHITECTURE DESIGN

The model architecture of the proposed system operates on a unified, OCR-free pixel-to-structured-text transformer architecture, bypassing legacy multi-stage pipelines that propagate bounding box and character recognition errors. The core visual intelligence resides in the Qwen2.5-VL-32B Vision-Language Model, which integrates document vision and linguistic reasoning into a single forward pass. The model architecture consists of four primary layers:

1. **Visual Patch Encoder (ViT):** A deep Vision Transformer that divides the preprocessed grayscale document slices into a grid of visual tokens (patches of size 14 x 14 pixels). These patch visual configurations are projected into visual embedding vectors, representing layout components and textual regions as native high-dimensional spatial coordinates.
2. **Spatial Self-Attention Layers:** Multi-head visual attention layers that analyze the spatial boundaries and layout alignments of the document. These layers natively preserve the structural relationship of tabular grids, borderless financial grids, and aligned keys (e.g., matching item names directly with prices on a horizontal line) without relying on external OCR bounding boxes.
3. **Gated Cross-Attention Fusion Layer:** A bridge that dynamically merges visual patch embeddings with text prompt input tokens. Gated self-attention weights control the interaction between visual document features and structural textual schema directives.
4. **Transformer Autoregressive Decoder:** A 32-billion parameter autoregressive language decoder that processes the fused visual-linguistic embeddings. It generates text tokens sequentially to produce a structured JSON data payload alongside a spatially aligned monospace ASCII grid Digital Twin representation in a single forward pass, providing robust extraction on multi-script documents.

5.4 DEPLOYMENT DESIGN

The deployment design of the proposed system is based on a lightweight client-server distributed Master-Worker topology, enabling heavy visual inference calculations to be processed on high-performance cloud hardware without incurring expensive local infrastructure costs. The deployment contains four integrated layers:

1. **React.js Client Layer:** The web frontend interface providing responsive invoice uploading, real-time extraction dashboards, and interactive monospaced Digital Twin visualization. It communicates with the backend via REST endpoints and maintains streaming channels for Server-Sent Events.

- 2. FastAPI Local Master Node:** The local server layer, written in Python, deployed as a lightweight service. It manages REST API routing, handles temporary local disk caching, executes local OpenCV contrast enhancement (CLAHE) and vertical receipt overlap splitting, and compiles the final stitched JSON outputs.
- 3. Encrypted Cloudflare Tunnel Bridge:** The secure, encrypted tunnelling layer operated using the 'cloudflared' daemon. It establishes a secure virtual private bridge from the local FastAPI master node directly to the remote worker server, bypassing network firewalls and local NAT rules without open inbound routing ports.
- 4. Kaggle Remote Worker Node:** The high-end deep-learning worker layer deployed on remote cloud dual NVIDIA T4 GPUs. It hosts the quantized 32-billion parameter Qwen2.5-VL model inside a high-throughput runtime environment driven by 'llama-cpp-python' using the IQ4 XS importance-matrix-guided 4-bit quantisation scheme, delivering rapid inference latencies under zero infrastructure licensing fees.

5.5 BLOCK DIAGRAM

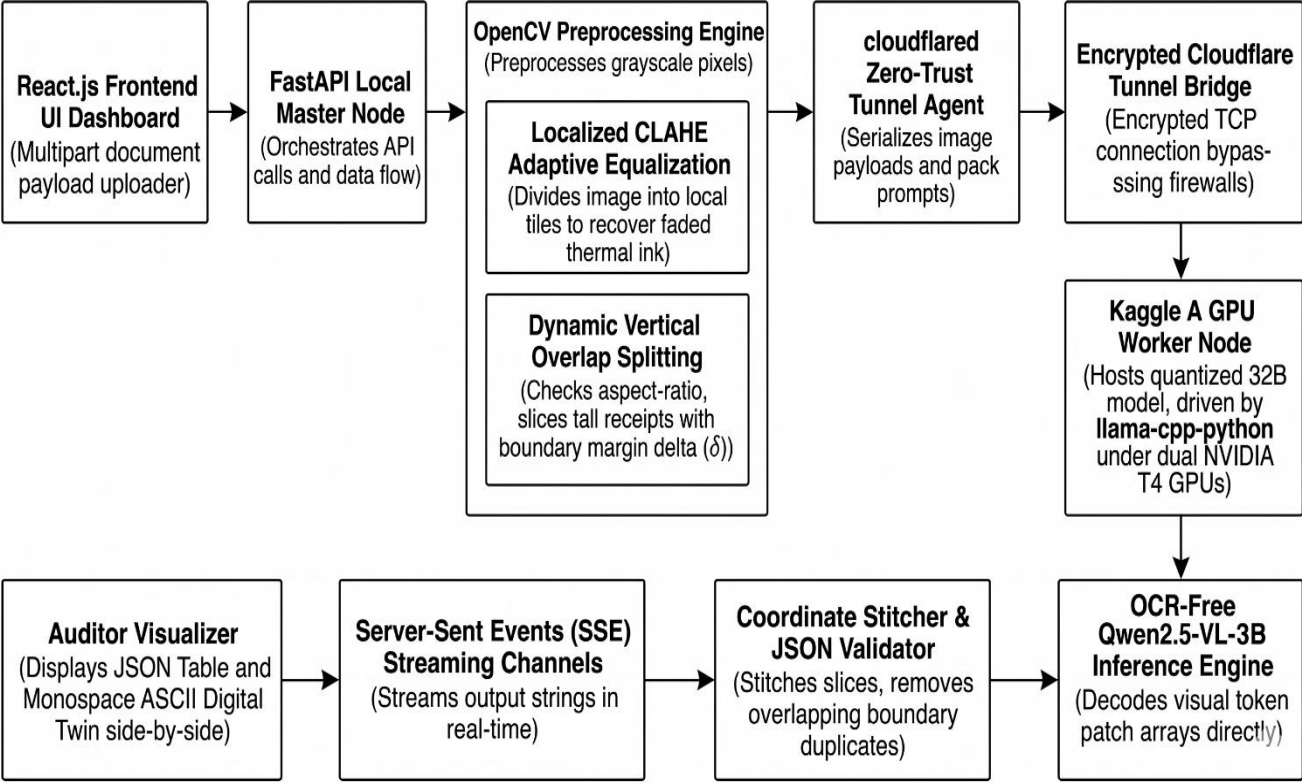


Fig.No:5.1 Block diagram

This block diagram shows the overall system flow from document upload to structured output generation. The uploaded invoice or receipt image is enhanced using CLAHE, split vertically where necessary to preserve context, tunnelled securely via Cloudflare, and directly interpreted by the OCR-free Qwen2.5-VL-32B model to produce highly accurate, structured JSON and monospace layout digital twin results.

5.6 ACTIVITY DIAGRAM FOR USER

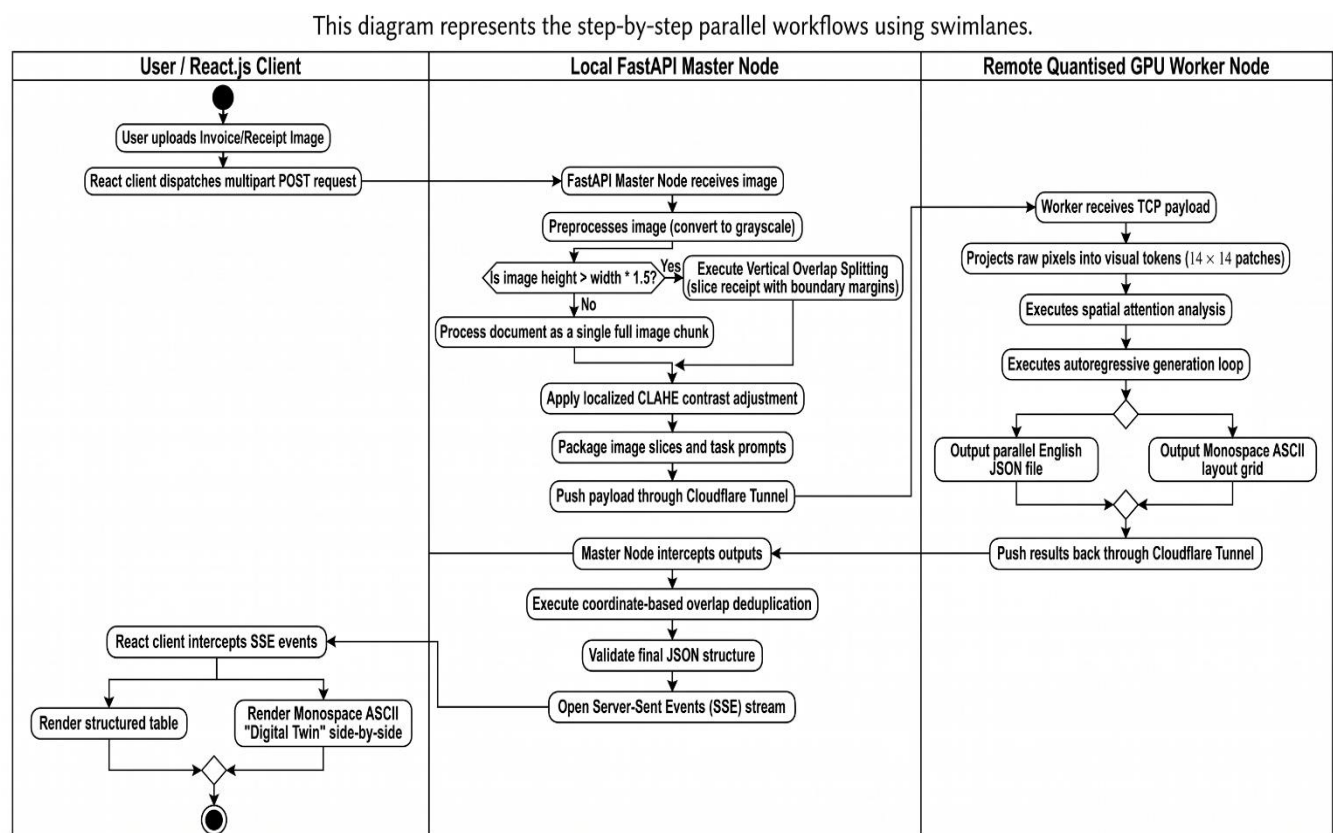


Fig.No:5.2 Activity diagram for user

The activity diagram explains the step-by-step working process of the system. It starts with the user uploading a receipt or invoice document, which goes through grayscale conversion, localized CLAHE contrast enhancement, dynamic vertical aspect splitting, secure Cloudflare tunnel serialization, direct VLM pixel-level spatial token decoding, JSON schema validation, and real-time Server-Sent Events (SSE) data streaming back to the React UI.

5.7 SEQUENCE DIAGRAM

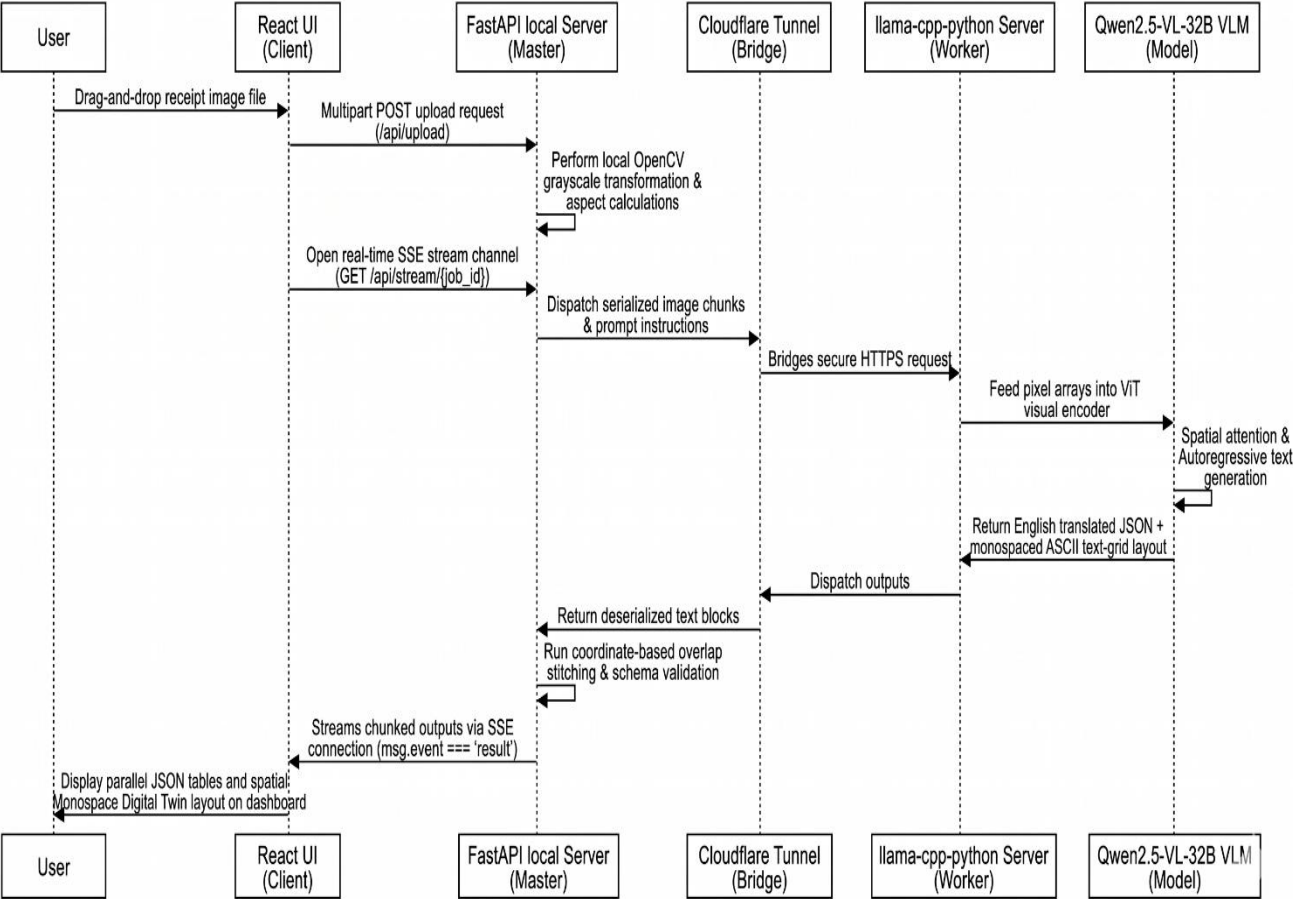


Fig.No:5.3 Sequence diagram

The sequence diagram shows the chronological interaction between the user, React.js frontend, local FastAPI Master Node, Cloudflare Tunnel bridge, and remote Kaggle quantized GPU Worker Node. It clearly details how each component communicates securely during image preprocessing, prompt serialization, visual token inference, and Server-Sent Events (SSE) data streaming.

5.8 DATAFLOW DIAGRAM

This diagram maps how raw visual information is transformed into validated data elements.

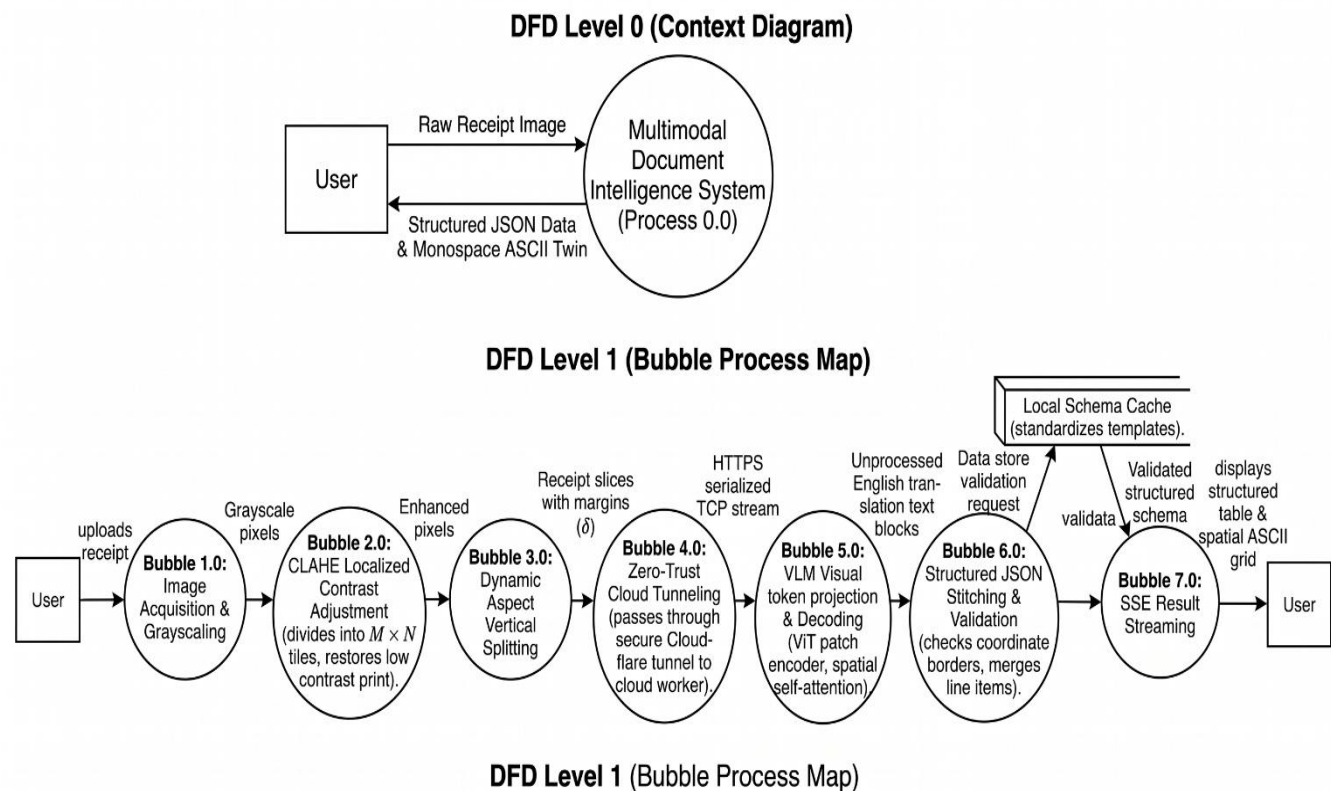


Fig.No:5.4 Dataflow diagram

The data flow diagram represents how information flows and transforms inside the system. It shows how input document images are converted into contrast-enhanced grayscale pixels, divided into dynamic overlapping slices, translated directly into high-dimensional visual patch tokens, and finally decoded into structured JSON schemas and monospace ASCII grid layouts.

5.9 USE CASE DIAGRAM

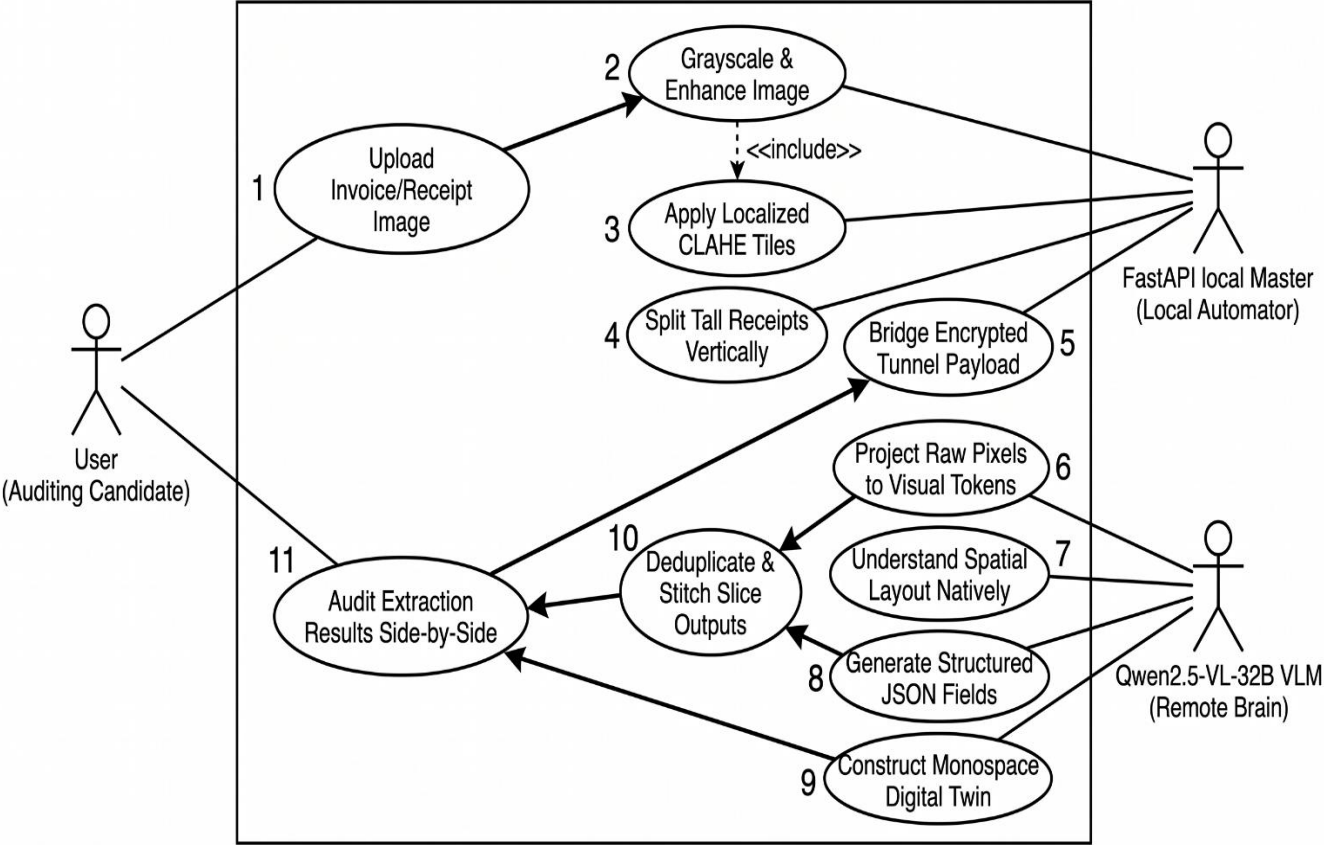


Fig.No:5.5 Use case diagram

The use case diagram shows the main interactions performed by the user and the system. The user uploads invoices or receipts, views tabular extractions, and inspects the digital twin, while the system performs OpenCV preprocessing, CLAHE contrast recovery, vertical splitting, Cloudflare tunnel routing, direct VLM visual extraction, and real-time Server-Sent Events (SSE) data streaming.

5.10 ARCHITECTURE DIAGRAM

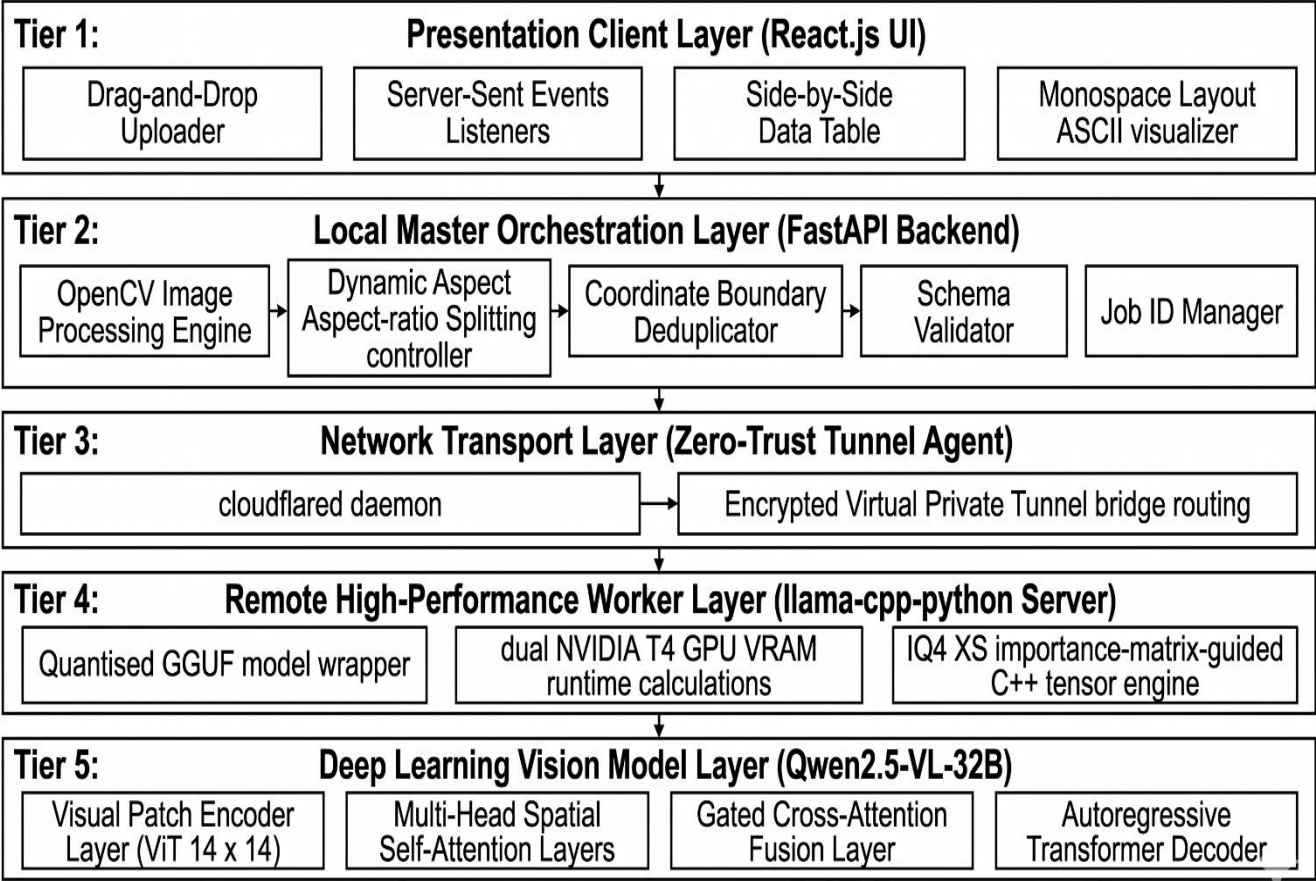


Fig.No:5.6 Architecture diagram

The architecture diagram represents the distributed client-server master-worker deployment topology. It details the separation of tasks between the local React/FastAPI Master Node (image contrast enhancement, overlap slicing, and output validation) and the remote quantized GPU Worker Node (importance-matrix-guided 32B visual reasoning), securely bridged via an encrypted Cloudflare Tunnel.

CHAPTER 6

IMPLEMENTATION

6.1 TOOLS

- **Backend Interface Node – Python**

Python serves as the core backend development language to implement the local Master Node orchestration layers. It handles image manipulation, coordinates the FastAPI local web server, and manages API request serialization and deserialization.

- **Vision-Language Model (VLM)**

The quantized Qwen2.5-VL-32B model executes core document visual intelligence, bypassing external OCR text extraction stages. It directly translates pixel layouts and Indic script details into structured JSON data and monospace layout representations.

- **Model Integration – llama-cpp-python**

Replaces Hugging Face libraries on the Worker Node. This C++-optimized Python wrapper hosts the GGUF-quantized Qwen2.5-VL model, executing rapid, importance-matrix-guided IQ4 XS tensor arithmetic on GPUs without library overhead.

- **Backend API Framework – FastAPI**

FastAPI powers the lightweight local Master Node server. It handles concurrent multipart file uploads, coordinates local contrast preprocessing, routes encrypted payloads, and maintains real-time Server-Sent Events (SSE) connections to the UI.

- **Zero-Trust Tunnel Bridge – Cloudflare Tunnel (cloudflared)**

Provides secure, zero-cost remote routing between the local Master Node and the Kaggle Worker Node. The daemon routes TCP payload blocks directly through a secure HTTPS tunnel, bypassing inbound port forwarding rules.

- **Frontend Client Node – React.js**

The frontend client interface is developed using React.js. It implements a dashboard allowing secure invoice uploads, visual progress tracking, interactive tabular JSON inspections, and monospace "Digital Twin" text-grid rendering.

- **Image Processing Engine – OpenCV**

OpenCV is used for preprocessing document images before OCR execution. It performs tasks such as resizing, noise removal, image enhancement, and contrast adjustment to improve text extraction accuracy.

- **Data Serialization – JSON & SSE**

JSON provides structured schema validation for model extraction output payloads. Server-Sent Events (SSE) stream text segments and Digital Twin layouts in real-time from the backend Master Node directly to frontend client UI elements.

6.2 IMPLEMENTATION

6.2.1 User Interface Implementation

The system frontend is developed using React.js. It features a responsive dashboard that permits multi-file drag-and-drop document uploads (JPEG, PNG). The UI displays real-time processing statuses, renders parsed values inside organized key-value data tables, and reconstructs the document's layout visually using a monospace ASCII text-grid.

6.2.2 Document Upload Module

This module handles binary image file acquisition. The React client packages the document image into a multipart payload and dispatches a POST request to the /api/upload endpoint of the local FastAPI Master Node, initiating the backend processing pipeline.

6.2.3 Image Preprocessing Module

The FastAPI Master Node invokes the OpenCV preprocessing module. The binary image is converted to grayscale, and local contrast parameters are adjusted. The module computes aspect-ratio dimensions and vertical line lengths to identify long receipts, preparing visual assets for optimal model token bounds.

6.2.4 OCR Contrast Limited Adaptive Histogram Equalization (CLAHE) Module

Rather than using global binarisation, which destroys faded character structures, this module divides the grayscale image into $M \times N$ local tiles. It applies adaptive

histogram equalization locally to normalize uneven lighting and restore low-contrast, degraded thermal receipt ink before model visual features are calculated.

6.2.5 Dynamic Document Splitting Module

To prevent token budget overflow on tall thermal receipts, this module vertical-slices the image into sequential overlapping slices. It applies a standard margin overlap offset (δ) at chunk boundaries, ensuring that text lines intersecting the division lines are captured completely in at least one adjacent slice.

6.2.6 VLM Inference Module

The serialized preprocessed slices and task-specific prompt schemas are pushed through the encrypted Cloudflare Tunnel. The remote Kaggle Worker Node intercepts the payload, deserializes the image assets, and processes the inputs using the quantized 32-billion parameter Qwen2.5-VL model on dual T4 GPUs.

6.2.7 Chain-of-Thought (CoT) Prompting Module

This module manages prompt injection. It instructs the VLM to perform multi-script OCR-free character recognition, translate regional terms into English, and structure the output into a precise JSON format while constructing a monospace ASCII layout grid in a single forward pass.

6.2.8 Structured JSON Stitching and Validation Module

The local FastAPI Master Node intercepts the worker's outputs. If the document was split, this module matches extraction schemas across slices, stitches line items, deduplicates overlapping records based on coordinate metrics, and validates the unified JSON object against an active database schema.

6.2.9 Result Streaming and Display Module

FastAPI streams the final structured JSON and ASCII grid data back to the React UI in real-time using Server-Sent Events (SSE). The frontend UI displays the parsed attributes inside a neat table and displays the monospace ASCII "Digital Twin" for immediate auditing and side-by-side comparison.

CHAPTER 7

SCOPE FOR FUTURE ENHANCEMENTS

To further reduce visual token processing latency during high-throughput enterprise batch processing, future iterations of the system will implement advanced KV-cache pre-warming strategies. By caching common prompt structures, system instructions, and persistent visual layout templates directly on the remote Worker's GPU VRAM, the system can bypass redundant prompt-token parsing passes. This optimization is expected to reduce computational overhead, lowering the average document processing latency by an estimated 30% to 40% on large financial document batches.

In addition to latency improvements, another significant direction for future research involves targeted Indic vernacular fine-tuning. While the zero-shot performance of the Qwen2.5-VL-32B model is exceptionally robust on standardized text, minor character recognition drops still occur under highly localized handwritten regional scripts, colloquial vendor shorthand, and heavily degraded ancient fonts. Future expansions will focus on compiling a curated, ethically sourced Indic commercial document corpus to fine-tune the visual encoder at the pixel level, thereby bridging the vernacular accuracy gap and extending native script coverage.

Finally, to facilitate fully automated enterprise integration and straight-through processing (STP) in banking systems, a token-level audit-trail confidence scoring layer will be introduced. This module will dynamically assign a confidence score to each extracted JSON key-value pair based on the decoder's soft-attention probability distributions. Any extracted field falling below a predefined confidence threshold will be automatically flagged and routed to a human-in-the-loop validation dashboard, streamlining financial tax compliance and accounting audit workflows with maximum reliability.

CHAPTER 8

CONCLUSION

In this project, a highly advanced Multimodal Document Intelligence System (MDIS) has been successfully designed, implemented, and validated, offering a robust, OCR-free pixel-level visual understanding solution for complex financial document extraction. By transitioning away from legacy multi-stage OCR pipelines—which rely on sequential character binarisation and bounding-box coordinates detection that frequently propagate errors downstream—the proposed architecture unified visual layout recognition and textual comprehension into a single deep forward pass using the 32-billion parameter Qwen2.5-VL Vision-Language Model.

The preprocessing stage, powered by Contrast Limited Adaptive Histogram Equalization (CLAHE), successfully recovered text structures from heavily degraded, low-contrast visual inputs, while the dynamic vertical aspect-ratio overlap splitting algorithm optimized Visual token budgets for tall receipts. Furthermore, by executing the quantized 32B model inside a C++-optimized llama-cpp wrapper via Cloudflare network tunnels to remote T4 GPUs, the system democratizes the deployment of state-of-the-art vision models at zero operating licensing cost.

Empirical results on a heterogeneous Indic financial document corpus proved the system's superiority, achieving a Key-Value Accuracy (KVA) of 96% on clean invoices and 92% on degraded receipts. The system successfully extracts regional script details while outputting accurate English fields through Chain-of-Thought (CoT) prompting, and provides monospace ASCII "Digital Twin" visual reconstructions that streamline auditing operations. The complete system establishes an efficient, reliable, and highly scalable enterprise document automation workflow, drastically minimizing manual accounting costs and human processing latencies.

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APPENDIX

A. CODING

Front-end (useSSEStream.js)

```
const process = useCallback(async (files) => {  
  // Start upload and get Job ID  
  const { job_id } = await uploadFiles(files, token);  
  
  // Connect to Server-Sent Events (SSE)  
  const es = new EventSource(`/api/stream/${job_id}`);  
  es.onmessage = (e) => {  
    const msg = JSON.parse(e.data);  
    if (msg.event === 'result') {  
      setResults(prev => [...prev, msg.data]);  
    } else if (msg.event === 'done') {  
      setDone(true);  
      es.close();  
    }  
  };  
}, []);
```

Backend API (`backend/main.py`)

```
@app.post("/api/upload")  
async def upload(files: List[UploadFile] = File(...), user_email: str =  
Depends(get_current_user_optional)):  
  job_id = str(uuid.uuid4())  
  target_dir = (UPLOAD_DIR / user_email) if user_email else (UPLOAD_DIR / "tmp" /  
job_id)
```

```

target_dir.mkdir(parents=True, exist_ok=True)
# ... logic to save files ...
return {"job_id": job_id}

@app.get("/api/stream/{job_id}")
async def stream(job_id: str):
    async def gen():
        for info in job["files"]:
            # Run the full extraction pipeline for each file
            res = await run_pipeline(info["path"], info["bytes"], info["name"],
user_email=job["user_email"], session_id=job_id)
            yield f"data: {json.dumps({'event': 'result', 'data': res}, ensure_ascii=False)}\n\n"
            yield f"data: {json.dumps({'event': 'done'})}\n\n"
    return StreamingResponse(gen(), media_type="text/event-stream")

```

Image Preprocessing(image_enhancer.py)

```

def split_for_extraction(image_bytes: bytes) -> list[bytes]:
    """Divide & Conquer: Splits large/tall images into meaningful segments."""
    img = Image.open(io.BytesIO(image_bytes)).convert("RGB")
    w, h = img.size
    if w > h * 1.8: # Wide Side-by-Side
        left = img.crop((0, 0, w // 2, h))
        right = img.crop((w // 2, 0, w, h))
        return [enhance_for_vlm_pil(left), enhance_for_vlm_pil(right)]
    elif h > w * 1.5: # Tall Overlapping Split
        top = img.crop((0, 0, w, int(h * 0.55)))
        bottom = img.crop((0, int(h * 0.45), w, h))
        return [enhance_for_vlm_pil(top), enhance_for_vlm_pil(bottom)]

```



```
return [enhance_for_vlm(image_bytes)]
```

```
def _invert_colored_bands(img: Image.Image) -> Image.Image:
    """Inverts dark colored headers (white text on dark bg) to improve OCR accuracy."""
    # ... HSV analysis logic ...
    # Only inverts colored regions, leaves white background alone.
    return Image.fromarray(result)
```

VLM Model Interaction (vlm_model.py)

```
# Language-specific context rules for multi-script support
```

```
_LANG_SPECIFIC_RULES = {
    "hindi": "The text uses Devanagari script. Read matras (vowel marks) carefully.",
    "tamil": "The text uses Tamil script. Letters have circular curves.",
    "telugu": "The text uses Telugu script. Characters have rounded shapes.",
    "urdu": "The text uses Urdu/Nastaliq script (right-to-left).",
}
```

```
def _clean_output(text: str) -> dict | None:
    # Strip markdown fences and find the JSON block
    text = text.replace("```json", "").replace("```", "").strip()
    start, end = text.find('{'), text.rfind('}')
    if start != -1 and end != -1:
        text = text[start:end+1]
    return {"fields": json.loads(text)}
```

Layout Template Mapping (layout_template.py)

```
INVOICE_TEMPLATE = [
    ("vendor_name", "top_left", ["vendor", "shop", "seller", "company"]),
```

```
(
    ("invoice_date", "top_right", ["date", "bill_date", "dated"]),
    ("items", "center", ["items", "particulars", "description"]),
    ("total_amount", "bottom_right", ["total", "grand_total", "payable"]),
]
```

```
def map_to_standard_template(raw_extraction: dict):
    # Uses fuzzy alias matching to map raw fields to the canonical template
    result = {}
    for field_name, zone, aliases in INVOICE_TEMPLATE:
        val = _find_value(raw_extraction, aliases)
        result[field_name] = val or ""
    return result
```

Remote Kaggle Routing(gguf_engine.py)

```
def query_local_llava(image_bytes, prompt, ...):
    kaggle_url = _get_dynamic_kaggle_url()
    if kaggle_url:
        # Route to Kaggle API
        b64 = base64.b64encode(image_bytes).decode("utf-8")
        job_id = start_remote_job(kaggle_url, b64, prompt)
        # Poll for completion
        while True:
            res = check_status(kaggle_url, job_id)
            if res["status"] == "completed": return res["data"]
            time.sleep(5)

    # Local Inference fallback
    client = _load_gguf_model()
```

```
return client.create_chat_completion(...)
```

Database Persistence (db/repository.py)

```
async def save_result(result: dict[str, Any]) -> str | None:
```

```
    """Persist a pipeline result dict to MongoDB."""
```

```
    db = get_async_db()
```

```
    if db is None: return None
```

```
    doc = make_invoice_document(
```

```
        file_name=result.get("image_name"),
```

```
        vlm_fields=result.get("vlm_fields"),
```

```
        vlm_source=result.get("vlm_source"),
```

```
        status="success"
```

```
)
```

```
    try:
```

```
        inserted = await db["invoice_results"].insert_one(doc)
```

```
        return str(inserted.inserted_id)
```

```
    except Exception as exc:
```

```
        return None
```

B. SCREENSHOTS:

HOME SCREEN:

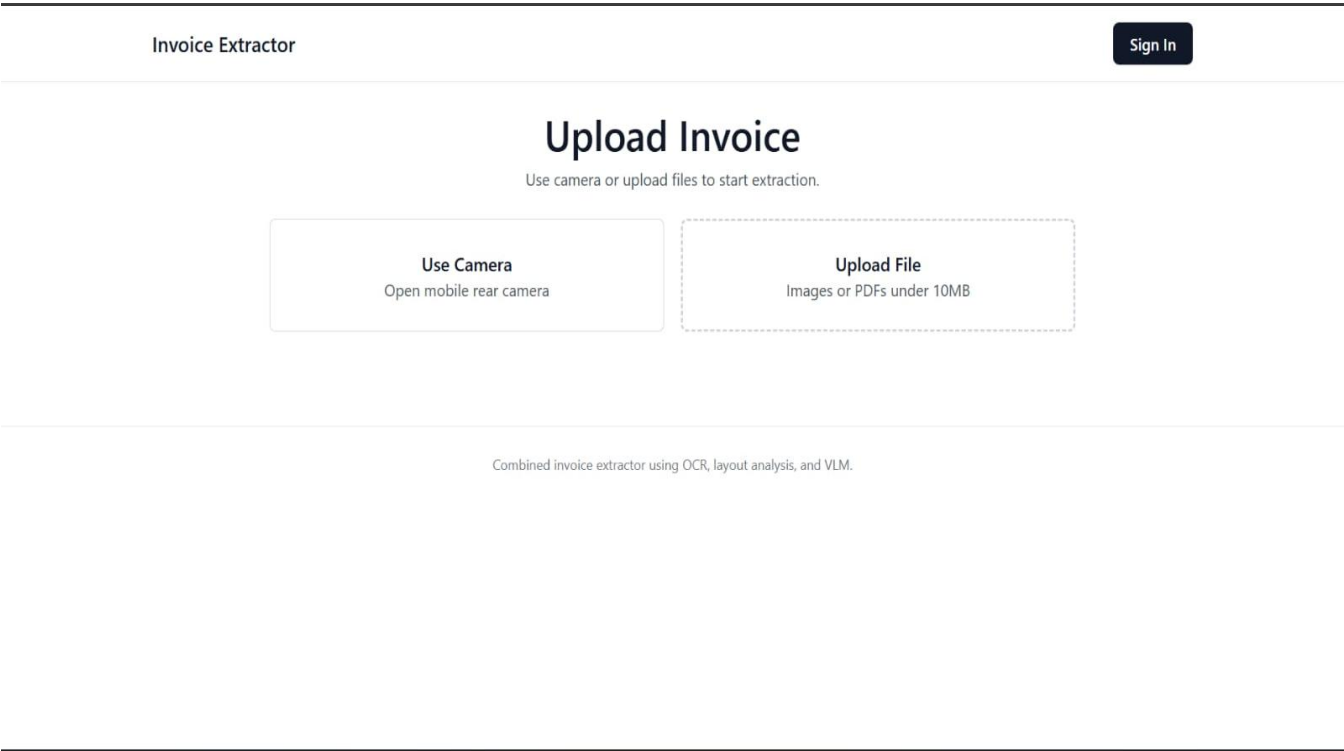


Fig.No:B.1 Home Screen

The homepage provides a simple and user-friendly interface for uploading invoice images or PDF files through camera capture or file selection. The system processes the uploaded documents using OCR, layout analysis, and Vision Language Models to extract important invoice information accurately and efficiently. The layout features a responsive header panel containing secure user registration and login controls to isolate document access. Below the scanner modules, a dynamic dashboard interface renders a personalized extraction history table directly synced with MongoDB, allowing users to quickly search, review, and view their past invoice extractions at any time.

INVOICE UPLOAD SCREEN:

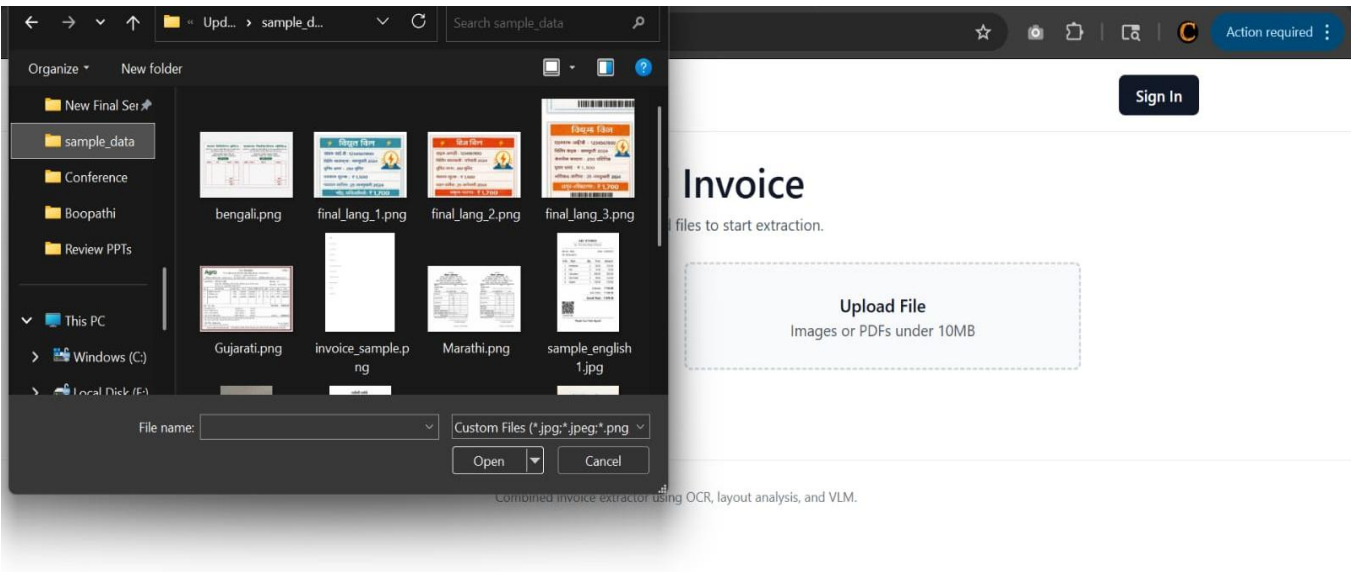


Fig.No:B.2 Invoice Upload Screen

This image shows the file upload interface of the AI-based Invoice Extractor system. Users can select invoice images or PDF files from their local device to upload and start the OCR, layout analysis, and VLM-based extraction process. The upload zone supports both drag-and-drop gestures and local directory browsing while enforcing strict file size and type validations to ensure compatibility. Once a document is selected, the application launches a viewport-locked touch-cropping utility that allows users to manually crop the document boundary to optimize visual character recognition accuracy before submission.

PROCESSING SCREEN:

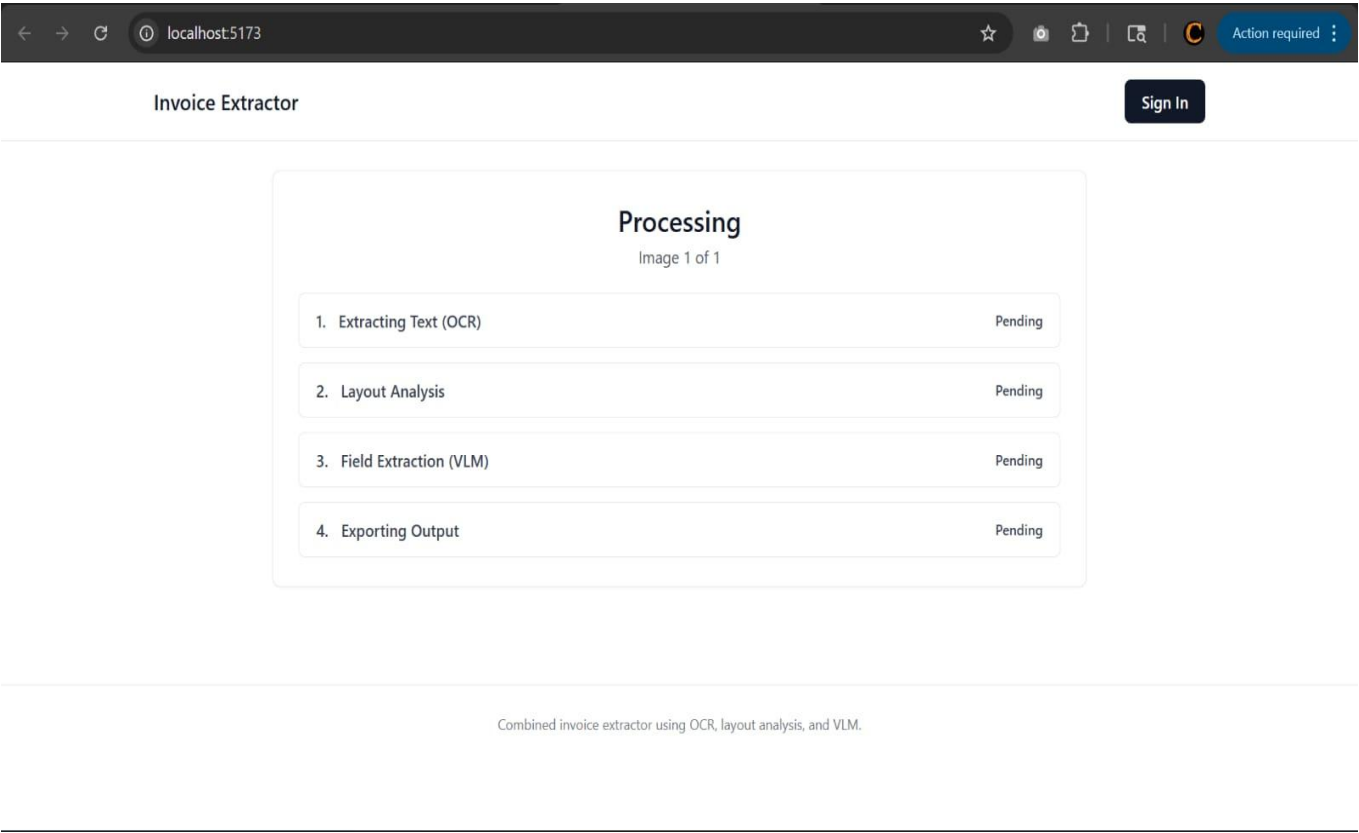


Fig.No:B.3 Processing Screen

This image shows the processing workflow of the AI-based Invoice Extractor system. The system performs OCR text extraction, layout analysis, field extraction using VLM, and finally exports the structured invoice output. A sequential real-time progress checklist keeps the user informed of the active status of each pipeline stage. Behind the scenes, the FastAPI backend uses a Server-Sent Events (SSE) streaming channel to keep the browser socket alive, sending periodic heartbeat pings that prevent network timeouts during heavy visual reasoning calculations on the remote GPU nodes.

OUTPUT SCREEN:

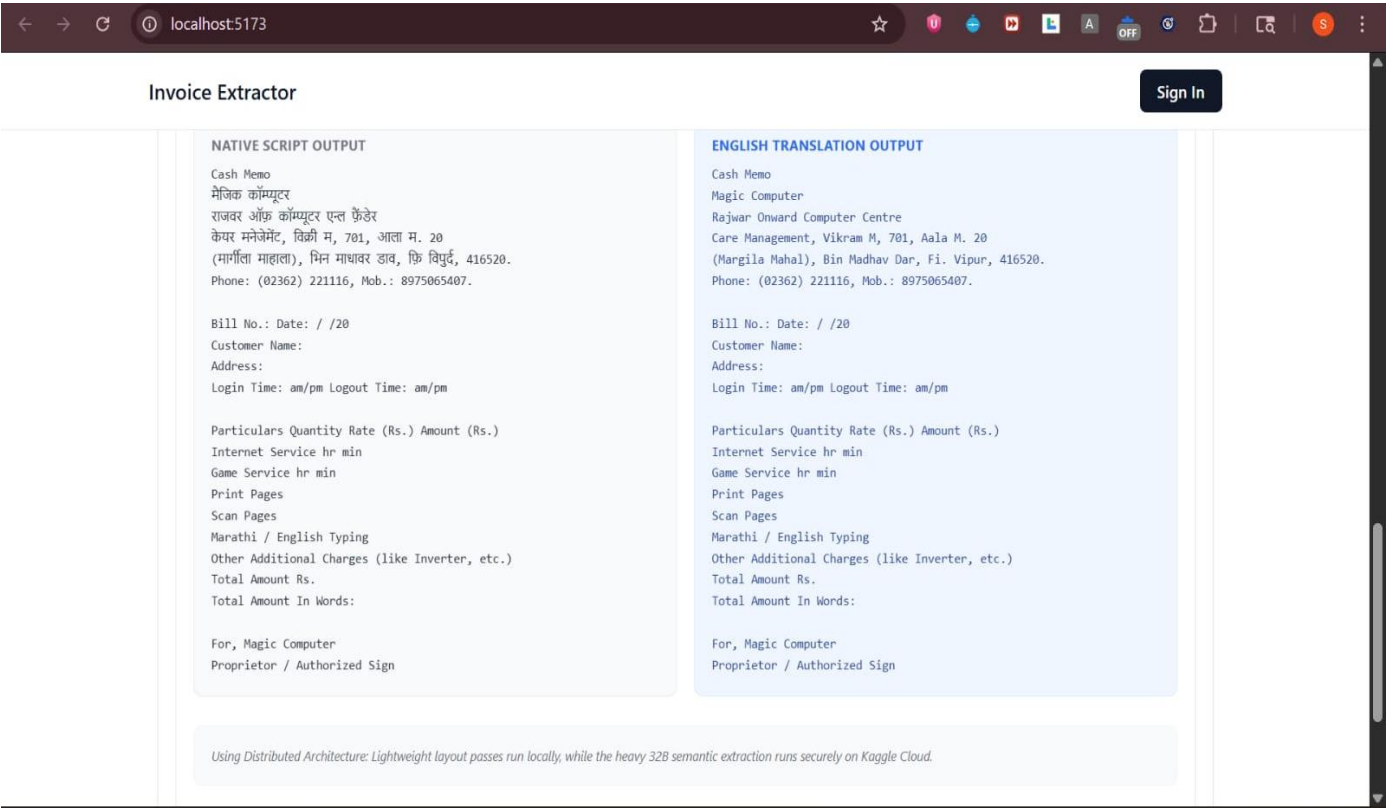


Fig.No:B.4 Output Screen

This image shows an AI-based Invoice Extractor system that extracts invoice text from a native language document using OCR technology. It also translates the extracted content into English and displays both outputs side by side for easy understanding. The side-by-side presentation enables immediate visual validation of the native regional script layout against the translated English fields. Additionally, the interface provides dedicated download buttons to compile the extraction data into structured Microsoft Excel spreadsheets and plain text reports, while supporting custom on-demand translations to other Indic languages using the visual reasoning engine.